

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (CL)

April 2012

CL301 Transportation Engineering-I

Date: 30.04.2012, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Enlist classification of roads and define each types of roads [05]
(b) State different factors controlling the highway alignment. Also draw neat sketch of obligatory points controlling the highway alignment. [02]
- Q - 2 (a) Write short note on Reconnaissance. [02]
(b) Derive equation for Extra Widening and explain mechanical and psychological widening. [06]
(c) Derive the equation for OSD. [06]

OR

- Q - 2 (a) Briefly describe the basic requirements of an ideal alignment. [02]
(b) Explain total reaction time of driver and the factors on which it depends. Also explain PIEV theory with neat sketch. [06]
(c) Calculate the length of transition curve and shift for $V = 65$ kmph, $R = 220$ m, rate of introduction of super elevation is 1 in 150, $W + W_e = 7.5$ m. Take $c = 0.57$ and $e = 0.07$ [06]
- Q - 3 (a) Explain penetration test with neat sketch [02]
(b) Enlist types of Traffic Signs. Draw at least 5 diagrams of each type of Traffic Signs. [06]
(c) Write a note on traffic flow characteristics and discuss the technical terms. [06]

OR

- Q - 3 (a) Discuss requirements of bitumen. [02]
(b) Write a note on Speed and Delayed Studies. Also discuss any two methods of speed and delayed studies. [06]
(c) Enlist the types of speed studies and discuss briefly spot speed study. [06]

SECTION – II

- Q - 4 (a) Write Two modes of construction of Cement Concrete Slab. Enlist Construction steps [04]

for Cement Concrete pavement Slab

(b) Define any 3 from the following: (Draw sketch wherever needed) [03]

1. Built Up Spray Grout
2. Penetration Macadam
3. Bituminous Surface Dressing
4. Seal Coat
5. Asphalt Concrete

Q - 5 (a) Enlist Classification of Joints and Explain any Two. [04]

(b) Design Flexible Pavement as per IRC guidelines for a new road using the following data. [10]

1. Two lane single carriageway
2. Initial Traffic in the year of completion of construction = 400 CVPD
3. Traffic Growth rate per annum = 8 %
4. Design Life = 15 years
5. Vehicle Damage Factor = 4
6. Design CBR of Sub grade Soil = 5 %

OR

Q - 5 (a) Enlist the construction procedure of earthen roads. [04]

(b) Design size and spacing of dowel bars at an expansion joint of concrete pavement of thickness 25 cm. Given the radius of relative stiffness of 80 cm. Design wheel load 5000 kg. Load capacity of the dowel system is 40 percent of design wheel load. Joint width is 2.0 cm and the permissible stress in shear, bending and bearing stress in dowel bars are 1000, 1400 and 100 kg/cm² respectively. [06]

(c) A cement concrete pavement of thickness 18 cm has two lanes of 7.2 m with a joint. Design the tie bars. [04]

Q - 6 (a) Define Surface and Sub-surface Drainage. The maximum quantity of water expected in longitudinal drains on clayey soil is 0.8m³/sec. Design the cross section and longitudinal slope of rectangular drain having bottom width of 1.0 m. The allowable velocity of flow in the drain is 1.5 m/sec and $n = 0.02$. [09]

(b) Write Various Sources of Revenue and Taxes levied by Governments in India [05]

OR

Q - 6 (a) The surface water from road side is drained to the longitudinal side drain across one half a bituminous pavement surface of total width 7.0 m, shoulder and adjoining land [09]

of width 8.0 m on one side of the drain. On the other side of the longitudinal drain, the water flows across from reserve land with grass and 2% cross slope towards the side drain, the width of this strip of land being 25 m. The run-off coefficients of the pavement, shoulder and reserve land with grass surface are 0.8, 0.25 and 0.35 respectively. The length of the stretch of land parallel to the road from where water is expected to flow to the side drain is about 400 m. Estimate the quantity of run-off flowing in the drain assuming 25 years period of frequency. Design the cross section and slope of the side drain in loamy soil with Manning's roughness coefficient = 0.022 and suitable speed of flow = 0.8 m/sec.

(b) Write a note on highway finance.

[05]

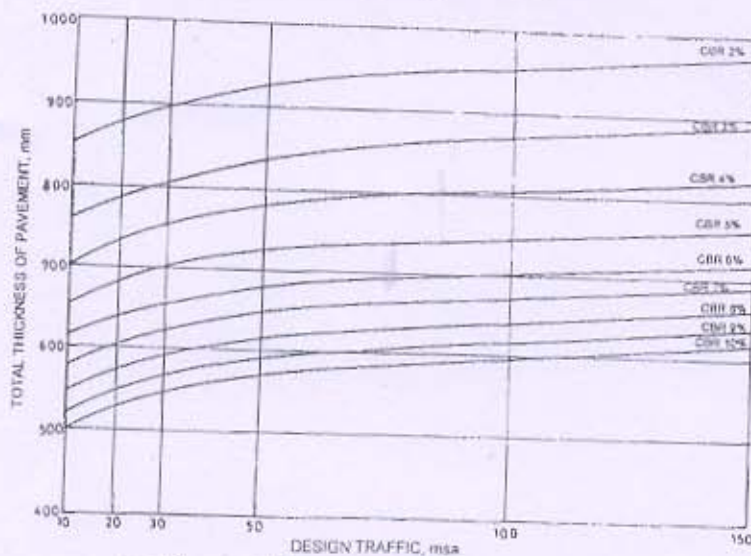


Fig. 2. Pavement Thickness Design Chart for Traffic 10-150 msa

PAVEMENT DESIGN CATALOGUE

PLATE 2 - RECOMMENDED DESIGNS FOR TRAFFIC RANGE 10-150 msa

Cumulative Traffic (msa)	Total Pavement Thickness (mm)	CBR 5%		
		PAVEMENT COMPOSITION		Granular Base & Sub-base (mm)
		Bituminous Surfacing		
		BC (mm)	DBM (mm)	
10	660	40	70	Base = 250
20	690	40	100	
30	710	40	120	
50	730	40	140	Sub-base = 300
100	750	50	150	
150	770	50	170	

